150mA, 75V Surface Mount Switching Diode

FEATURES

- Low power loss, high efficiency
- High surge current capability
- Hermetically sealed glass
- RoHS Compliant

APPLICATIONS

- Switching mode power supply (SMPS)
- Adapters
- Lighting application
- On-board DC/DC converter

MECHANICAL DATA

- Case: MMELF
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Polarity: Indicated by cathode band
- Weight: 30.60mg (approximately)

KEY PARAMETERS		
PARAMETER	**VALUE**	**UNIT**
I_F	150	mA
V_{RRM}	75	V
I_{FSM}	4	A
V_F at I_F = 100mA	1	V
$T_{J\,MAX}$	175	°C
Package	MMELF	
Configuration	Single die	

Pb-Free RoHS COMPLIANT

MMELF

ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)				
PARAMETER		**SYMBOL**	**VALUE**	**UNIT**
Power dissipation		P_D	500	mW
Repetitive peak reverse voltage		V_{RRM}	75	V
Forward current		I_F	150	mA
Non-repetitive peak forward surge current	t = 1s	I_{FSM}	0.5	A
	t = 1ms		1.0	A
	t = 1μs		4.0	A
Repetitive peak forward current		I_{FRM}	450	mA
Junction temperature range		T_J	-65 to +175	°C
Storage temperature range		T_{STG}	-65 to +175	°C

THERMAL PERFORMANCE			
PARAMETER	**SYMBOL**	**LIMIT**	**UNIT**
Junction-to-ambient thermal resistance	$R_{\Theta JA}$	300	°C/W

ELECTRICAL SPECIFICATIONS (T_A = 25°C unless otherwise noted)

PARAMETER		CONDITIONS	SYMBOL	MIN	MAX	UNIT
Reverse Breakdown Voltage		I_R = 100μA	V_{BR}	100	-	V
		I_R = 5μA		75	-	V
Forward voltage[1]	LL4448 LL914B	I_F = 5mA, T_J = 25°C	V_F	0.62	0.72	V
	LL4148	I_F = 50mA, T_J = 25°C		-	1	V
	LL4448 LL914B	I_F = 100mA, T_J = 25°C		-	1	V
Reverse current @ rated V_R[2]		V_R = 20V, T_J = 25°C	I_R	-	25	nA
		V_R = 75V, T_J = 25°C		-	5	μA
Junction capacitance		1 MHz, V_R=0V	C_J	-	4	pF
Reverse Recovery Time		I_F = I_R = 10mA, R_L = 100Ω, I_{RR} = 1mA	t_{rr}	-	4	ns

Notes:

1. Pulse test with PW = 0.3ms
2. Pulse test with PW = 30ms

ORDERING INFORMATION

ORDERING CODE[1]	PACKAGE	PACKING
LL4148 L0	MMELF	10,000 / 13" Tape & Reel
LL4448 L0	MMELF	10,000 / 13" Tape & Reel
LL914B L0	MMELF	10,000 / 13" Tape & Reel
LL4148 L0G	MMELF	10,000 / 13" Tape & Reel
LL4448 L0G	MMELF	10,000 / 13" Tape & Reel
LL914B L0G	MMELF	10,000 / 13" Tape & Reel

Notes:

1. Above ordering codes L0/L0G refer to physically identical parts without any differences

CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.1 Typical Forward Characteristics

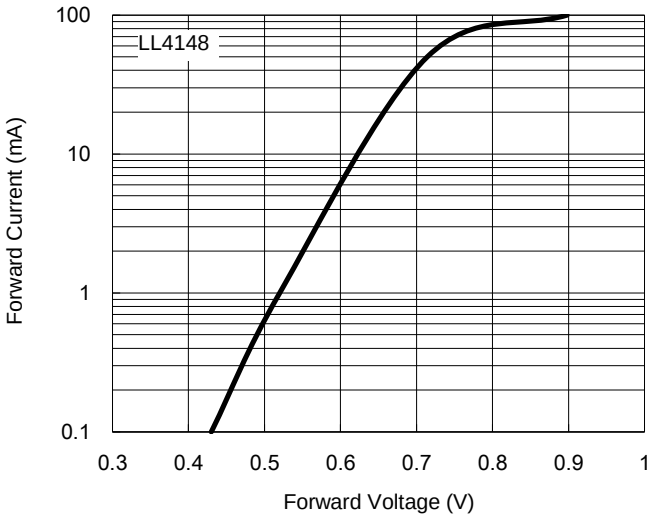


Fig.2 Reverse Current VS. Reverse Voltage

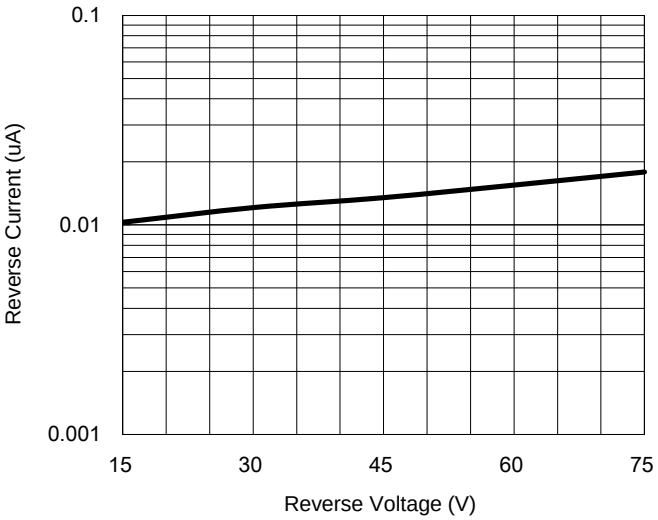


Fig.3 Admissible Power Dissipation Curve

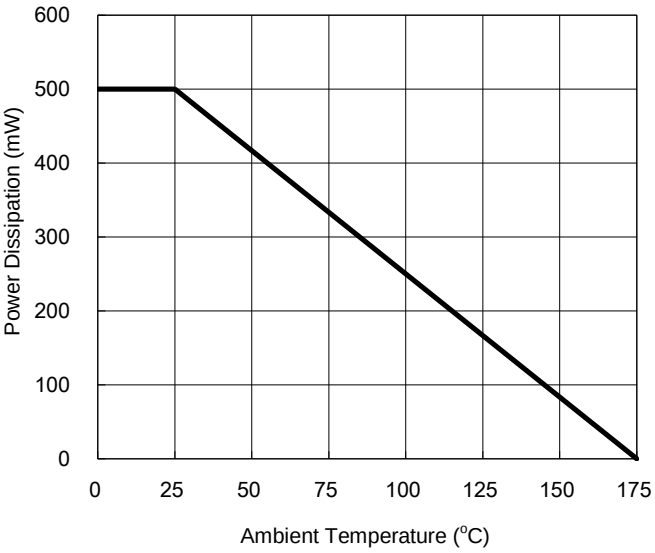
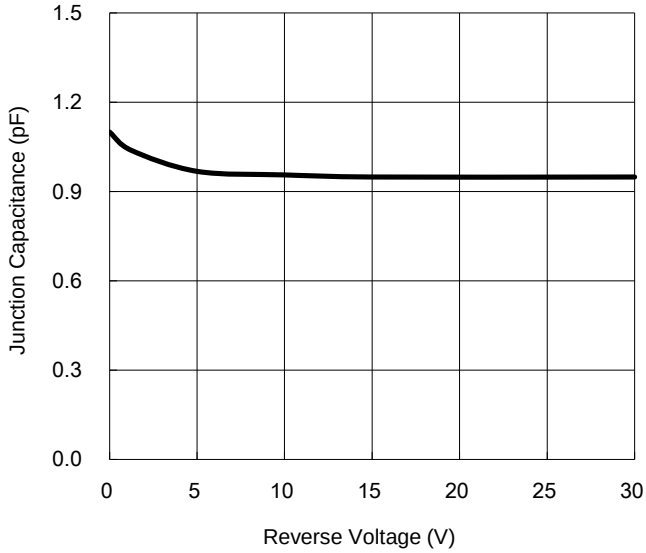


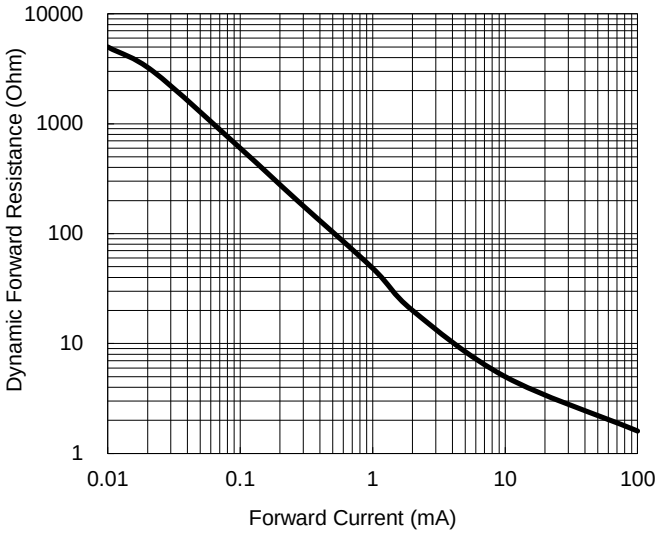
Fig.4 Typical Junction Capacitance



CHARACTERISTICS CURVES

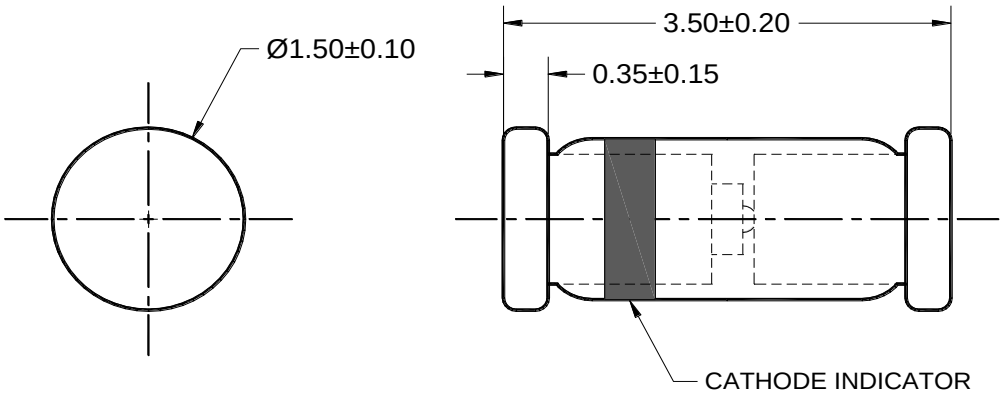
(T_A = 25°C unless otherwise noted)

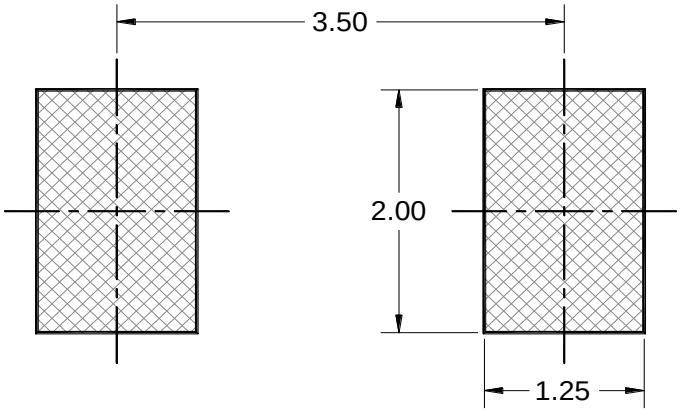
Fig.5 Forward Resistance VS. Forward Current



PACKAGE OUTLINE DIMENSIONS

MMELF





SUGGESTED PAD LAYOUT

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.
3. PACKAGE OUTLINE REFERENCE: JEDEC DO-213, VARIATION AA, ISSUE D.
4. DWG NO. REF: HQ2SD07-MMELFG-044 REV A.

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